Dhairyya Singh
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ACADEMIC POSITIONS

Postdoctoral Research Associate, Princeton University **2026-Present**
Princeton Computational Memory Lab
PI: Dr. Kenneth Norman

Research Specialist, University of Pennsylvania **2019-2020**
Penn Computational Cognitive Neuroscience Lab
PI: Dr. Anna Schapiro

Teaching Fellow, Ashoka University **2018-2019**
Department of Psychology
PI: Dr. Bittu Rajaraman

EDUCATION

University of Pennsylvania **Spring 2021-**
PhD., Psychology **Spring 2026**
Interdisciplinary Training in Computational Neuroscience (ITCN)
PI: Dr. Anna Schapiro
Committee: Dr. Sudeep Bhatia & Dr. Sharon Thompson-Schill

Ashoka University **2017-2018**
Post-Graduate Diploma in Advanced Studies and Research (DipASR) – Psychology (Advanced Major) and Computer Science (Minor)
Magna cum Laude

Ashoka University **2014-2017**
B.A., Psychology (Major)
Magna cum Laude

HONORS AND GRANTS

Society for Neuroscience Trainee Professional Development Award (TPDA) **2021**

IBRO-Simons Computational Neuroscience Imbizo Summer School Attendee, Cape Town, South Africa **2019**

NCBS Computational Approaches to Memory and Plasticity (CAMP) Summer School Attendee, Bangalore, India **2019**

Department of Psychology, Best Undergraduate Thesis Award **2018**

Dean of Faculty and Research Travel Grant **2018**

Dean's List
For both DipASR semesters **2017-2018**
For all six B.A. semesters **2014-2017**

Dean of Faculty and Research Travel Grant **2017**

Department of Psychology, Builder's Award **2017**

APA Division Society for Personality and Social Psychology Summer Program for Undergraduate Research (SPUR) Scholar **2016**

PUBLICATIONS & PREPRINTS

Singh, D.*, Dong, C.*, Tandoc., M. C. & Schapiro, A. C. (in press). Statistical learning drives predictive shifts in object memory. *Journal of Experimental Psychology: General*. https://doi.org/10.31234/osf.io/yuxb6_v1

* Equal contribution

Singh, D., Schapiro, A. C. (2026). Evidence for complementary learning systems within the hippocampus. *Philosophical Transactions of the Royal Society B: Biological Sciences*.

Zhou, Z., **Singh, D.**, Tandoc, M. C., & Schapiro, A. C. (2023). Building integrated representations through interleaved learning. *Journal of Experimental Psychology: General*, 152(9), 2666–2684. https://doi.org/10.1037/xge0001415

Singh, D., Norman, K. A., & Schapiro, A. C. (2022). A model of autonomous interactions between hippocampus and neocortex driving sleep-dependent memory consolidation. *Proceedings of the National Academy of Sciences*, 119(44), e2123432119. https://doi.org/10.1073/pnas.2123432119

Majeed, N. A., **Singh, D.**, et al. (2022). Symbolic quantitative cognition in wild zebrafish (Danio rerio). *bioRxiv*. https://doi.org/10.1101/2022.03.17.484678

Singh, D., & Chatterjee, G. (2017). The evolution of religious belief in humans: A brief review with focus on cognition. *Journal of Genetics*, *96*(3), 517-524. https://doi.org/10.1007/s12041-017-0794-7

PRESENTATIONS

Singh, D., Norman, K.A., & Schapiro, A.C. (2026, November 14-18). *A mixture-of-experts model of hippocampal-mPFC interactions* [Conference presentation]. Society for Neuroscience, Washington DC, United States.

Singh, D., Williams. A.B., & Schapiro, A.C. (2026, August 3-6). *A mixture-of-experts model of hippocampal-mPFC interactions*. [Poster presentation]. Cognitive Computational Neuroscience, New York City, NY, United States.

Singh, D., Williams. A.B., & Schapiro, A.C. (2026, May 28-29). *A mixture-of-experts model of hippocampal-mPFC interactions*. [Poster presentation]. Context and Episodic Memory Symposium, Philadelphia, PA, United States.

Singh, D., Williams. A.B., & Schapiro, A.C. (2025, August 12-15). *Resolving the tension between exemplar and structure learning through a mixture-of-experts model of hippocampal-PFC interactions*. [Poster presentation]. Cognitive Computational Neuroscience, Amsterdam, The Netherlands.

Singh, D.*, Dong, C.*, Tandoc, M. & Schapiro, A.C. (2024, April 13-16). *Statistical learning drives predictive shifts in object memory*. [Symposium presentation]. Cognitive Neuroscience Society, Toronto, ON, Canada.

Singh, D.*, Dong, C.*, Tandoc., M. & Schapiro, A.C. (2022, November 12-16). *Predictive shifts in object representations with statistical learning*. [Poster presentation]. Society for Neuroscience, San Diego, CA, United States.

Singh, D., Norman, K.A., & Schapiro, A.C. (2022, May 12-13). *A model of autonomous interactions between hippocampus and neocortex driving sleep-dependent memory consolidation.* [Conference presentation]. Context and Episodic Memory Symposium, Philadelphia, PA, United States.

Dong, C.*, **Singh, D.***, Tandoc., M. & Schapiro, A.C. (2022, May 12-13). *Predictive shifts in object representations with statistical learning*. [Poster presentation]. Context and Episodic Memory Symposium, Philadelphia, PA, United States.

Singh, D., Norman, K.A., & Schapiro, A.C. (2021, November 8-11). *A neural network model of sleep-dependent consolidation via autonomous hippocampal-cortical replay* [Conference presentation]. Society for Neuroscience, Online.

Singh, D., & Rajaraman, K. (2018, July). *Simple numerical cognition in zebrafish (Danio rerio)*. [Poster presentation]. International Congress of Neuroethology, Brisbane, Australia.

INVITED TALKS

Princeton Parallel Distributed Processing (PDP) Group	**Fall 2026**
Lab meeting, Favila Lab, Brown University	**Spring 2026**
CNI +/-	**Spring 2025**
Penn Chronobiology and Sleep Institute Research Meeting	**Spring 2023**
Penn Memory Seminar	**Fall, 2022**
Lab meeting, Norman Lab, Princeton University	**Spring 2022**

GUEST LECTURES

PSYC3300 Seminar in Sleep and Memory **Spring, 2023**
Topic: "*Computational Models of Sleep*"
Instructor: Prof. Anna Schapiro

PSYC429 Seminar in Sleep and Memory **Spring, 2022**
Topic: "*Computational Models of Sleep*"
Instructor: Prof. Anna Schapiro

MENTORSHIP

Cody V. Dong (Penn Undergraduate)	**Spring 2021 –**
MindCORE Summer Fellowship winner	**Summer 2022**
Jason Lim (Penn Undergraduate)	**Spring 2023 –**
Greenberg Fellowship winner	**Fall 2023**

TEACHING EXPERIENCE

Teaching Assistant, PSYC1530-401 Memory **Fall, 2022**
Instructor: Prof. Anna Schapiro; 100 Students (Penn)

Teaching Assistant, PSYC149-401 Cognitive Neuroscience **Spring, 2022**
Instructor: Prof. Michael Arcaro; 105 Students (Penn)

Teaching Fellow, BIO205/PSY211 Introduction to Neuroscience **Spring, 2019**
Instructor: Prof. Kaveri Rajaraman; 35 Students (Ashoka)

Teaching Fellow, PSY202 Statistics and Research Methodology II **Spring, 2019**
Instructor: Prof. Rashmi Nayyar; 75 Students (Ashoka)

Teaching Fellow, BIO205/PSY211 Introduction to Neuroscience **Fall, 2018**
Instructor: Prof. Kaveri Rajaraman; 90 Students (Ashoka)

Teaching Assistant, PSY202 Statistics and Research Methodology II **Spring, 2018**
Instructor: Prof. Kai Qin Chan; 45 Students (Ashoka)

AD HOC REVIEWING

Nature; eLife; Science Advances; Journal of Cognitive Neuroscience; PLOS One, Open Mind; PLOS Computational Biology

SERVICE

Princeton Neuroscience Institute Seminar Committee	**2026**
MindCORE Representative, DEEPenn STEM Resource Fair	**2025**
Data and Outcomes Committee Member, Penn MindCORE Mind DivE In	**2024, 2025**
Prospective Graduate Student Panel, Dept. of Psychology	**2024, 2025**
PhD Application Support, Neuroscience Graduate Group	**2023**
Graduate Student Representative, Dept. of Psychology	**2022-23**

Journal Club Leader, Penn High School Neuroscience Research Academy	**2022, 2023, 2024, 2025**
Mentor, Penn MindCORE Mind DivE In	**2021, 2023, 2025**